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Original Article

Hyperforin-loaded gold nanoparticle alleviates experimental autoimmune encephalomyelitis by suppressing Th1 and Th17 cells and upregulating regulatory T cells

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Abstract

Hyperforin an herbal compound, is commonly used in traditional medicine due to its anti-inflammatory activities. The aim of this study was to use a hyperforin loaded gold nanoparticle (Hyp-GNP) in the treatment of experimental autoimmune encephalomyelitis (EAE) an animal model of multiple sclerosis (MS). Hyp-GNP and hyperforin significantly reduced clinical severity of EAE, which was accompanied by a decrease in the number of inflammatory cell infiltration in the spinal cord. Additionally, treatment with Hyp-GNP significantly inhibited disease-associated cytokines as well as an increase in the anti-inflammatory cytokines in comparison to all groups including the free-hyp group. Furthermore, hyperforin and Hyp-GNP inhibited the differentiation of Th1 and Th17 cells while promoting Treg and Th2 cell differentiation via regulating their master transcription factors. The current study demonstrated the although, free-hyp improved clinical and laboratory data Hyp-GNP is significantly more efficient than free hyperforin in the treatment of EAE.
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Key words: Hyperforin; Multiple sclerosis; Experimental autoimmune encephalomyelitis (EAE); Myelin oligodendrocyte glycoprotein (MOG); Gold nanoparticle

Multiple sclerosis (MS) is a neurodegenerative autoimmune disease of the central nervous system (CNS).[1] To date, the etiology of MS is still unknown, but abnormal response of the immune system to self-myelin antigens accompanied by a defect in immunoregulatory network, plays a major role in disease pathogenesis.[2]

EAE is the best experimental model for the study of the immunological and pathological mechanisms in human MS.[3]

Both MS and EAE have long been considered as a Th1-mediated autoimmune disease.[4] The Th1/Th2 paradigm has provided the basis for understanding the mechanisms involved in the pathogenesis of different autoimmune diseases.[5]

Th1 and Th2 differentiation is organized by the transcription factors T-bet and GATA3, respectively.[6,7] IFN-γ and expression of T-bet prevent Th2 differentiation, whereas IL-4 and GATA3 expression antagonizes Th1 polarization.[8] However, autoreactive Th1 cell responses and Th1/Th2 balance, could not fully explain the pathogenesis of MS and EAE.[9] Recently some studies have demonstrated that Th17 cells play an important pathogenic role in inflammatory and autoimmune responses in a variety of human autoimmune diseases.[10]

Autoimmune encephalomyelitis may be promoted not only by aberrant pathogenic T cell responses, but also by dysfunction or impaired maturation of Treg cells.[11]

Further evidence demonstrates that the initiation and progression of EAE are associated with the production and activity of pro-inflammatory cytokines.[12] Th1 and Th17 cells have the ability to orchestrate the influx of inflammatory cells into the CNS lesions through the secretion of pro-inflammatory cytokines.[4] In contrast, anti-inflammatory cytokines[13,14] may have protective effects against the progression of disease in EAE animals.

Despite intensive research, treatment of MS has yet remained unknown. The development of nontoxic and safe medications from medicinal herbs needs to be considered for disease management.[15]

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There is increasing knowledge about the pharmacological activities of hyperforin (a constituent of St. John's wort) to explore its potential in the disease treatment.[16] Recent studies have shown that hyperforin has antidepressant and anti-inflammatory[17] as well as the antioxidant activity.[18] Thus, it seems that hyperforin has a therapeutic potential in MS treatment.

Current pharmacotherapy for EAE/MS, including β-interferon and glatiramer acetate, has been proven ineffective due to a long treatment period, several side effects, high cost and also is non-effective in some patients, hence, it appears that new medications need to be discovered. As mentioned in the previous sentences, based on the anti-inflammatory effects of hyperforin, it can be considered as a candidate for treatment of inflammatory based diseases including MS. Additionally, one of the main limitation of the current MS therapies is very low accumulation of the drug into the disease site and long-term pharmacotherapy leads to significant inefficacy and severe toxicity.[19] In order to overcome the challenge, we proposed a new formulation of hyperforin bound to the GNP (Hyp-GNP) that serves as drug carrier.[19]

Gold nanoparticles (GNP) are a good candidate because of their ability to enter various cell types with high efficiency,[20] biocompatibility, low level of cytotoxicity[21] and high loading efficiency.

Therefore, the aim of this study was to evaluate the effects of hyperforin in free form and conjugated with GNP on the immunological properties of EAE.

Methods

Animals

Female C57BL/6 mice (aged 8-10 weeks) were purchased from Pasteur Institute of Iran. The animals were maintained at a constant temperature with a 12 hour light/dark cycle and had free access to water and pellet diet. Experiments were conducted in accordance with the NIH *Guide for the Care and Use of Laboratory Animals*, with approval from the Animal Ethic Committee of Mashhad University of Medical Sciences.

Synthesis of gold nanoparticle products

Synthesis of gold nanoparticle (GNP)

Synthesis of GNP was accomplished with citrate reduction reaction using a modified Frens method.[22] In detail, 10 mg tetrachloroauric acid (III) trihydrate (HAuCL4.3H2O, Sigma-Aldrich, St. Louis, MO, USA) was added to 25 ml of deionized water to make 1 mM HAuCL4 solution. This solution was placed on a hot plate with a magnetic stirrer and boiled until the temperature reached 95 °C. Then 4 ml of trisodium citrate (38.8 mM) was added to the solution quickly. The mixtures were continuously heated until the solution turned ruby red. This red color indicates the formation of stable gold nanoparticles. The gold nanoparticles were filtered using 2.5 μm filters and stored at 4 °C for further use.

Synthesis of GNP-hyperforin conjugate (Hyp-GNP) and drug loading efficiency

Ninety milliliters of GNP (1 mM) was taken in a flask and was placed on a magnetic stirrer. Then 15 mg of hyperforin (Phtoplan-Diehm & Neuberger-GmbH, Heidelberg, Germany) was added to 30 ml of solvent to make 1 mM hyperforin solution. This solution was added to AuNP solution and stirred for 2 hours. The change in the color of solution from ruby red to purple indicated the formation of Hyp-GNP.

The mixtures were then centrifuged at 14,000 g and 4 °C for 30 min to separate unadsorbed hyperforin from Hyp-GNP conjugates. The pellet was re-suspended in Milli-Q water and washed three times. Unconjugated hyperforin was measured by HPLC method. The amount of adsorbed hyperforin calculated from the difference between the initial and the free hyperforin concentrations. The amount of hyperforin adsorbed onto GNP was calculated to be 9.97 mg (66.5%).

Characterization of gold nanoparticle products

The shape, size and morphology of gold nanoparticles were characterized by UV–visible spectrophotometer, Zetasizer Nano ZS and atomic force microscopy (AFM).

UV–visible spectroscopy

To characterize the synthesized gold nanoparticles, they were analyzed by an ultraviolet–visible (UV–vis) spectrophotometer (UV-1700, Shimadzu, Kyoto, Japan). For sample preparation 1 ml aliquots of each GNP were diluted in 3 ml distilled water. All samples were loaded into a 1 cm path length quartz cuvette for UV–vis spectrometry readings and the spectrum of the reactions was recorded within a range of 400-700 nm.

Particle size analysis

The average particle size and zeta potential of nanoparticles were determined by dynamic light scattering, DLS (Malvern Zetasizer Nano-ZS, Malvern Instruments Ltd., UK), with a laser beam at a wavelength of 633.8 nm. Each sample was analyzed in triplicate at 20 °C at a scattering angle of 173°. Zeta potential data were collected through electrophoretic light scattering at 20 °C, 150 V, in triplicate in pure water. The instrument was calibrated with Malvern 50 V standard.

Atomic force microscopy

Samples of the gold nanoparticles products were centrifuged and redispersed in deionized water. Samples were passed through a 0.22 μm filter, and an aliquot of reaction mixture was dried in a grade III hood on a clean glass slide for 2 hours. Finally, the samples were analyzed using an intermittent contact mode atomic force microscope (JPK Instruments AG-Berlin, Germany), placed on top of an inverted optical microscope (Olympus IX81, Tokyo, Japan).

Drug release assay and in vitro cytotoxicity

The protocols of drug release assay and in vitro cytotoxicity and their related results are presented in supplementary file.

Induction and evaluation of EAE

For EAE induction, the animals were immunized with subcutaneous injection in the flank of an emulsion of $MOG_{35\text{-}55}$ (SBS Genetech Co. Ltd., Beijing, China) and complete Freund's adjuvant (H37Ra strain; Sigma, St. Louis, MO). Mice were also injected intraperitoneally with 250 ng pertussis toxin (Sigma-Aldrich, St. Louis, MO) on the day of induction and 2 days later.

The clinical symptoms were scored daily as depicted in Table 1. The mice were weighed and observed for clinical score

Table 1
Grading system for clinical assessment of EAE.

Score	Clinical signs
0	No symptoms
1	Partial loss of tail tonicity
2	Complete loss of tail tonicity
3	Flaccid tail and abnormal gait
4	Hind leg paralysis
5	Hind legs paralysis with hind body paresis
6	Hind and foreleg paralysis
7	Death

and confirmed by a blinded observer. Then mice were assessed for the onset day of disease, maximum mean clinical score (MMCS), incidence, and cumulative disease index.

Study design

Mice were randomly divided into six groups, and 8 animals in each group. The first group (control) was EAE induced and received PBS intraperitoneally (i.p.). In the second group, EAE was induced and received free-GNP (1 mM) (i.p.). In the treatment groups, three doses of hyperforin-GNP conjugate, 1, 5 and 10 mg/kg, were administered to groups 3, 4 and 5 (low, intermediate and high doses of Hyp-GNP), respectively via i.p. injection. These groups were named Hyp-GNP1, Hyp-GNP2 and Hyp-GNP3, respectively. The amount of GNP in hyperforin-GNP conjugate was equivalent to free-GNP (1 mM).

Free hyperforin (free-Hyp) at a dose of 10 mg/kg/mice was administered to group 6 (i.p.). The dose of hyperforin in free form or conjugated with GNP was based on preliminary dose-finding experiments in our previous study.[23]

All animals in the study were treated daily from day 0 until day 21 post immunization. At the end of the experimental period (21 days), the mice were sacrificed as per animal ethics.

Histopathology

On day 21 after EAE induction, the mice were anesthetized and their spleens were removed before undergoing intra-cardiac perfusion with PBS containing 4% paraformaldehyde and then the mice were sacrificed. Paraffin embedded sections (5 μm thick) of the spinal cords were obtained and stained with hematoxylin and eosin (H&E) to assess leukocyte infiltration. Slides were assessed in a blinded manner (by analysis of three fields) and histological scores were determined as our previous study.[23]

Detection of regulatory T cells by flow cytometry

The effect of Hyp-GNP products, and free-Hyp on Treg cells was determined by flow cytometry technique using mouse regulatory T cell staining kit (eBioscience, San Diego, CA).

Splenocytes were isolated from all treatment groups and for detection of cell surface markers, cells (1×10^6) were washed and stained in staining buffer with the following fluorochrome labeled monoclonal antibodies: fluorescein isothiocyanate (FITC)-conjugated rat anti-mouse CD4 (clone RM4) and phycoerythrin (PE)-conjugated rat anti-mouse CD25 (clone PC61.5) antibodies at 4 °C for 30 min.

Table 2
The sequences of primer and probes which are used in the study.

Genes	Sequences	
Mouse Foxp3	F	5′-CAgAgTTCTTCCACAACA-3′
	R	5′-CATgCgAgTAAACCAATg-3′
	Probe	5′-FAM TgAgTgTCCTCTgCCTCTCCg TAMRA-3′
Mouse GATA3	F	5′- CTgCggACTCTACCATAA-3′
	R	5′-gTggTggTCTgACAgTTC-3′
	Probe	5′-FAM CTgCTCTCCTTgCTgCCgAC TAMRA-3′
Mouse T-bet	F	5′-TgTggTCCAAgTTCAACC-3′
	R	5′-CATCCTgTAATggCTTgTg-3′
	Probe	5′-FAM TCATCACTAAgCAAggACggCg TAMRA-3′
Mouse ROR-γt	F	5′-ggATgAgATTgCCCTCTA-3′
	R	5′-CCTTgTCgATgAgTCTTg-3′
	Probe	5′-FAM CTCATCAATgCCAACCgTCCTg TAMRA-3′
B2mG	F	5′- CCTgTATgCTATCCAgAA-3′
	R	5′-gTAgCAgTTCAgTATgTTC-3′
	Probe	5′-FAM TATACTCACgCCACCCACCg TAMRA-3′

For detection of Foxp3, the flow cytometry technique was described in our previous study.[23]

Quantitative real-time PCR (qPCR)

The effects of Hyp-GNP products, and hyperforin on mRNA expression of transcription factors (Foxp3, T-bet, ROR-γt and GATA3) were determined by qPCR. The brain and spleen cells were isolated from EAE mice treated with hyperforin and Hyp-GNPs. Total RNA was isolated using Trizol reagent (Roche Diagnostics GmbH, Mannheim, Germany), in accordance with the manufacturer's protocol. cDNA was synthesized using Prime-ScriptTM^{RT} reagent kit (Takara Biotechnology, Otsu-Shiga, Japan) according to the manufacturer's instructions. Quantitative real time PCR was performed on the Rotor Gene Q (Qiagen Hilden, Germany) using Taqman Master Mix (Prime mix EXTaq™, Takara Biotechnology, Otsu-Shiga, Japan) with appropriate primers and probe (Table 2). Mouse β2 micro globulin (β2mG) was used as an endogenous control for sample normalization. Based on expression of target genes normalized to β2mG, we calculated the relative quantification delta-delta *C*t and the results are presented as fold change compared to control.

Cytokine enzyme-linked immunosorbent assay (ELISA)

To determine the in vitro response, spleens were isolated from EAE mice on day 21-post immunization. Mononuclear cells from spleens of immunized mice were prepared and cultured in 24 well plates in RPMI-1640 medium at a concentration of 2×10^6 cells/well with 10% FCS, 100 U/ml penicillin, and 100 μg/ml streptomycin in the presence of 20 μg/ml MOG. After 72 hours, supernatants were collected and cytokine contents were detected using ELISA kits for IL-4, IL-6, IL-10, IL-17A, IFN-γ and TGF-β (eBioscience, San Diego, CA, USA) according to the manufacturer's protocol.

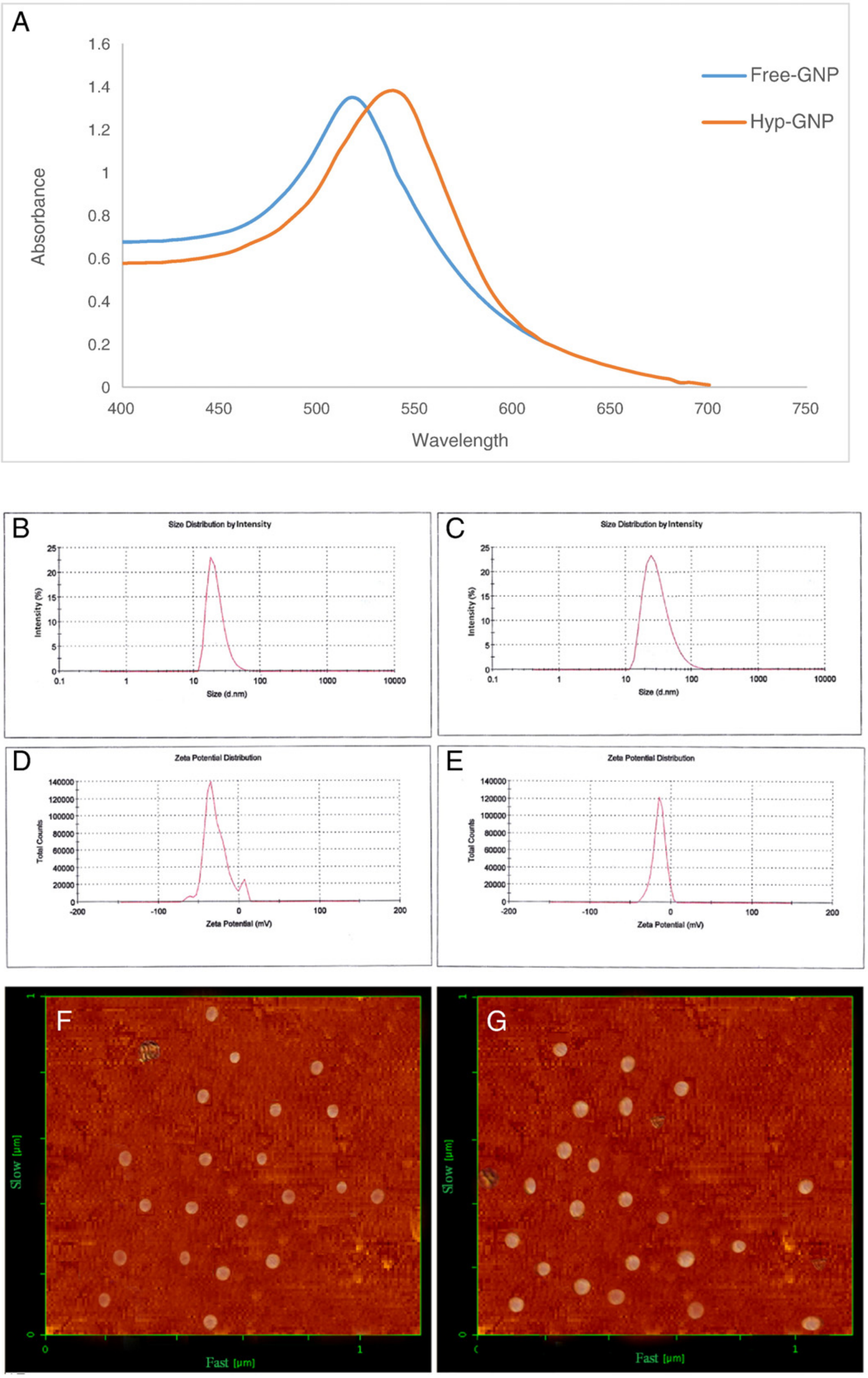

Figure 1. The characterization of GNPs. **(A)** UV–vis spectra of gold nanoparticles synthesized using HAuCL4 with sodium citrate solution (free-GNP, solid line) and after hyperforin loading onto the gold nanoparticle surface (Hyp-GNP, dash line). The absorbance intensity of GNPs is peaked at 517 and 537 nm, respectively. **(B and C)** Average particle size obtained from DLS data for **(B)** free-GNP and **(C)** Hyp-GNP. The sizes of free-GNP and Hyp-GNP are estimated to be 22.16 and 33.96 nm, respectively. **(D and E)** Zeta potential measurements of free-GNP and Hyp-GNP, respectively. **(F)** AFM images of GNPs synthesized using HAuCL4 with sodium citrate solution (free-GNP) and **(G)** after hyperforin loading onto the gold nanoparticle surface (Hyp-GNP).

Table 3
Physical characterization of gold nanoparticle products.

Nanoparticle type	Mean particle size (nm)	Polydispersity index	Zeta potential (mV)
Free-GNP	22.16 ± 0.2	0.05 ± 0.023	−30.2 ± 1.2
Hyp-GNP	33.96 ± 0.35	0.065 ± 0.018	−14.5 ± 1.4

Data analysis

All data were reported as mean ± SEM. Differences between groups were measured using one-way analysis of variance (ANOVA) followed by the Tukey post hoc test. Statistical evaluation was analyzed using SPSS 19 (IBM SPSS, Armonk, NY). Statistical significance was defined as $P < 0.05$ (*), $P < 0.01$ (**), $P < 0.001$ (***).

Results

Characterization of GNPs products

UV–vis spectroscopic analysis of gold nanoparticles

The UV–vis spectrum of free-GNP clearly indicates the characteristic sharp absorbance peak at 517 nm (Figure 1, *A*, solid line) corresponding to the surface plasmon resonance (SPR) band of the spherical GNPs, confirming the synthesis of small gold nanoparticles as observed by other studies.[24]

UV–vis scanning of GNP products showed that the absorbance intensities of free-GNP and Hyp-GNP peaked at 517 and 537 nm, respectively. The absorbance intensity showed that as hypeforin was coated onto the GNPs the SPR maximum of gold nanoparticles (λ_{max}) shifted to a higher wavelength from 517to 537 nm due to increased size.

Dynamic light scattering measurements

According to the DLS results, the average hydrodynamic diameters of free-GNP and Hyp-GNP products were found to be 22.16 ± 0.2 and 33.96 ± 0.35 nm, respectively (Figure 1, *B* and *C*). The observed size patterns are in good agreement with the data provided by ultraviolet spectroscopy and atomic force microscopy.

Table 3, also depicts the magnitude of zeta potentials of the particles. The zeta potential is −30.2 ± 1.2 mV for free-GNP (Figure 1, *D*). This provided enough surface charge to stabilize the particles from aggregation. Zeta potential was reduced to −14.5 ± 1.4 mV when hyperforin was conjugated onto the surface of GNP (Figure 1, *E*). Although the surface charge is reduced to some extent, Hyp-GNPs have enough repulsive force to prevent from aggregation during long-term storage.

Atomic force microscopy analysis of gold nanoparticle products

The structural properties of free and Hyp-coated GNPs were analyzed using AFM microscopy.

Figure 1, *F* and *G* shows a typical AFM image of free-GNPs and Hyp-GNPs. AFM analysis of these particles showed an average diameter of 20 ± 2 and 30 ± 1.9 nm, respectively. These results were comparable to the results obtained by DLS.

The images showed that the GNPs are approximately in a spherical shape and only a few particles with triangular shapes could be identified.

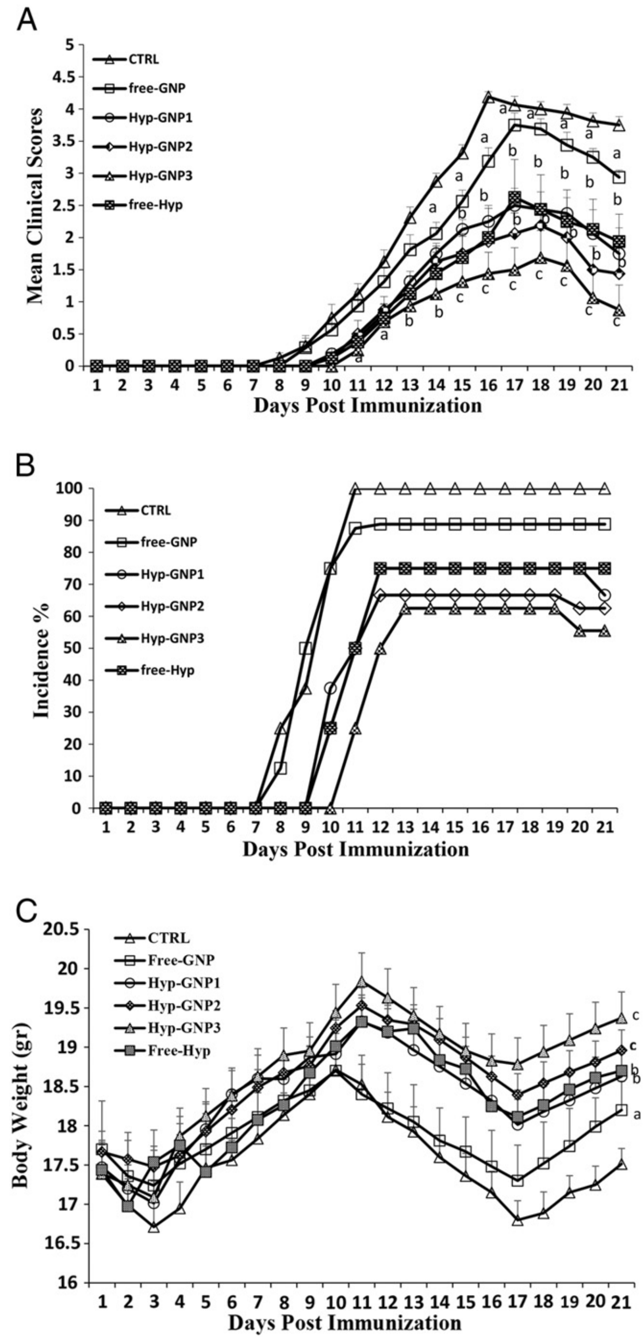

Figure 2. Clinical symptoms of EAE after treatment with Hyp-GNPs and hyperforin. The figure shows that Hyp-GNPs and hyperforin ameliorate clinical symptoms of EAE. Mice were monitored for signs of EAE, and the results were presented as **(A)** mean clinical scores **(B)** incidence of disease and **(C)** mean body weight.[a]$P < 0.05$, [b]$P < 0.01$, [c]$P < 0.001$ compared with vehicle treated control.Values are presented as mean ± SEM.

Hyp-GNPs and free-Hyp improve the clinical course of EAE in C57BL/6 mice

As shown in Figure 2, all vehicle-treated mice developed EAE, and displayed the first signs of disease onset with tail weakness on day 9.6 ± 0.4 reaching a maximum mean clinical score (MMCS) of 4.56 ± 0.11 on day 17 (Figure 2, *A*). In

Table 4
Clinical features of EAE in the administration of Hyp-GNP and hyperforin.

Group	Incidence (sick/total)	Mean day of onset	Maximum mean clinical score	Cumulative disease index (CDI)[a]
1—Vehicle	8/8 (100%)	9.62 ± 0.42	4.18 ± 0.09	36.18 ± 1.28
2—Free-GNP	8/9 (88.8%)	10.12 ± 0.39	3.75 ± 0.18*	29.75 ± 1.9*
3—Hyp-GNP1	6/8 (75%)	10.87 ± 0.29	2.5 ± 0.26**	20 ± 2.19**
4—Hyp-GNP2	6/9 (66.6%)	11.14 ± 0.29	2.18 ± 0.34**	17.18 ± 2.8**
5—Hyp-GNP3	5/8 (62.5%)	12 ± 0.37	1.68 ± 0.42***	12.43 ± 3.2**
6—Free-Hyp	6/8 (75%)	11 ± 0.3	2.62 ± 0.58**	18.87 ± 4.2**

Values are given as the mean ± SEM. CDI, cumulative disease index.
[a] Sum of clinical scores over the entire period.
* $P < 0.05$ versus the control.
** $P < 0.01$ versus the control.
*** $P < 0.001$ versus the control.

contrast, when mice were administered free-Hyp, disease incidence was reduced to 75% (Figure 2, *B*) and disease severity was significantly reduced, with MMCS of 2.62 ± 0.58 on day 17 (MMCS: vehicle group 4.56 ± 0.11 vs. free hyp-treated mice 2.62 ± 0.58; $P < 0.01$) (Figure 2, *A*, Table 4).

In addition, in mice receiving Hyp-GNP3 with an equal dose to free-Hyp, disease incidence was reduced to 62.5% (Figure 2, *B*) and MMCS was reduced to 1.68 ± 0.4 (Figure 2, *A*), on day 18, suggesting a milder disease progress compared with control or hyperforin alone. (MMCS, vehicle group 4.56 ± 0.11 vs. Hyp-GNP1, 2.5 ± 0.26 Hyp-GNP2, 2.18 ± 0.34; $P < 0.05$, and Hyp-GNP3 treated mice 1.68 ± 0.4; $P < 0.01$). Hyp-GNP3 treatment delayed the onset of disease (day 12), indicating that Hyp-GNP treatment significantly delayed the development of EAE when hyperforin was provided in the nanoparticle form. Figure 2, *C* illustrates that, such as EAE scores and incidence, weight loss was more improved after treatment with Hyp-GNP3 than equal dose of free-Hyp.

Hyp-GNPs and free-Hyp treatment decreased infiltration of inflammatory cells into the CNS

Figure 3, *A* shows that spinal cords from vehicle-treated mice had massive leukocyte infiltration with several foci of inflammation.

In contrast, in spinal cords from hyperforin-treated mice, the numbers of inflammatory cells were reduced, compared to vehicle treated EAE mice.

Analysis of Hyp-GNP treated mice resulted in a marked reduction in infiltration of inflammatory cells in the spinal cord and only showed mild signs of inflammation, confirming the clinical observation that these mice are largely protected against EAE (Figure 3, *A* and *B*).

Analysis of Treg cell population in Hyp-GNP and free-Hyp treated mice

Analysis of Treg cell population in free-Hyp treatment group indicated an increased number of $CD4^+CD25^+FoxP3^+$ Tregs, compared to vehicle-treated mice (Figure 4, *A*). As shows in Figure 4, *A* and *B*, the difference in the percentages of Tregs (7.27 ± 0.24 versus 10.86 ± 0.34) between the two groups reached statistical significance ($P < 0.01$).

Treatment with 1, 5 and 10 mg/kg Hyp-GNP led to a dose-dependent increase in $CD4^+CD25^+FoxP3^+$ Tregs numbers (Figure 4). The increase was most significant in the 10 mg/kg Hyp-GNP group (Hyp-GNP3), where about 2-fold increase (7.27 ± 0.24% in vehicle vs.13.97 ± 0.28% in 10 mg/kg Hyp-GNP-treated group; $P < 0.001$) was observed.

The results revealed that, while the number of Tregs in hyperforin and Hyp-GNP were around 10 and 14%, only 8.15% regulatory cells were found in free-GNP treated mice (Figure 4) indicating that free-GNP showed no effect on Treg cell number compared to control group (7.27 ± 0.24% in vehicle vs. 8.15 ± 0.3% in free-GNP-treated group; $P > 0.05$) while both free-Hyp and Hyp-GNPs significantly promoted Treg expansion and/or induction.

Hyp-GNPs and free-Hyp decrease pro-inflammatory and increase anti-inflammatory cytokines

In vitro analysis of supernatants collected from MOG stimulated splenocytes showed significantly decreased secretion of the pro-inflammatory cytokines IFN-γ, IL-17A and IL-6 from cells in free-Hyp treated mice, in comparison with control mice (Figure 5, *A*, *B* and *C*, $P < 0.001$ for IFN-γ and IL-17A and $P < 0.05$ for IL-6). In contrast, the release of Th2/Treg cytokines, TGF-β, IL-10 andIL-4, was highly enhanced in spleen cell cultures from free-Hyp treated animals (Figure 5, *D*, *E* and *F*, $P < 0.01$).

Similar results were also achieved from splenocytes isolated from free-GNP and Hyp-GNPs treated mice. The results showed that treatment with Hyp-GNP1, Hyp-GNP2 and Hyp-GNP3 dose dependently inhibited IFN-γ, IL-17A and IL-6 ($P < 0.001$ for IFN-γ and IL-17A and $P < 0.01$ for IL-6) and increased TGF-β, IL-10 and IL-4 in comparison with vehicle treated mice ($P < 0.001$). Treatment with free-GNP inhibited IFN-γ, IL-17A and IL-6 and increased IL-4 and IL-10 but not TGF-β compare with vehicle treated mice ($P < 0.05$ for IFN-γ, IL-6 and IL-4 and $P < 0.01$ for IL-17A and IL-10).

Hyp-GNP treatment also showed a significant effect on the TGF-β, IL-10 and IL-4 levels as compared with free-GNP and free-Hyp ($P < 0.05$).

Effects of free-hyp and Hyp-GNPs on the polarization of $CD4^+$ T cells in EAE mice

Figure 6 shows free-Hyp treated mice displayed a significant decrease in the transcription of ROR-γt and T-bet in the spleen (Figure 6, *C* and *D*) and brain (Figure 6, *G* and *H*) compared to vehicle treated mice ($P < 0.01$). In addition, free-Hyp treated mice showed elevated expression of Foxp3 and GATA3 (Figure 6, *A* and *B*) in the spleen and brain (Figure 6, *E* and *F*), while vehicle treated mice had decreased expression of Foxp3 and GATA3 ($P < 0.01$).

Furthermore, treatment with free-GNP and Hyp-GNPs resulted in reduced expression of both T-bet and ROR-γt and increased expression of GATA3 but the expression of Foxp3 increased in Hyp-GNP treated groups (Figure 6).

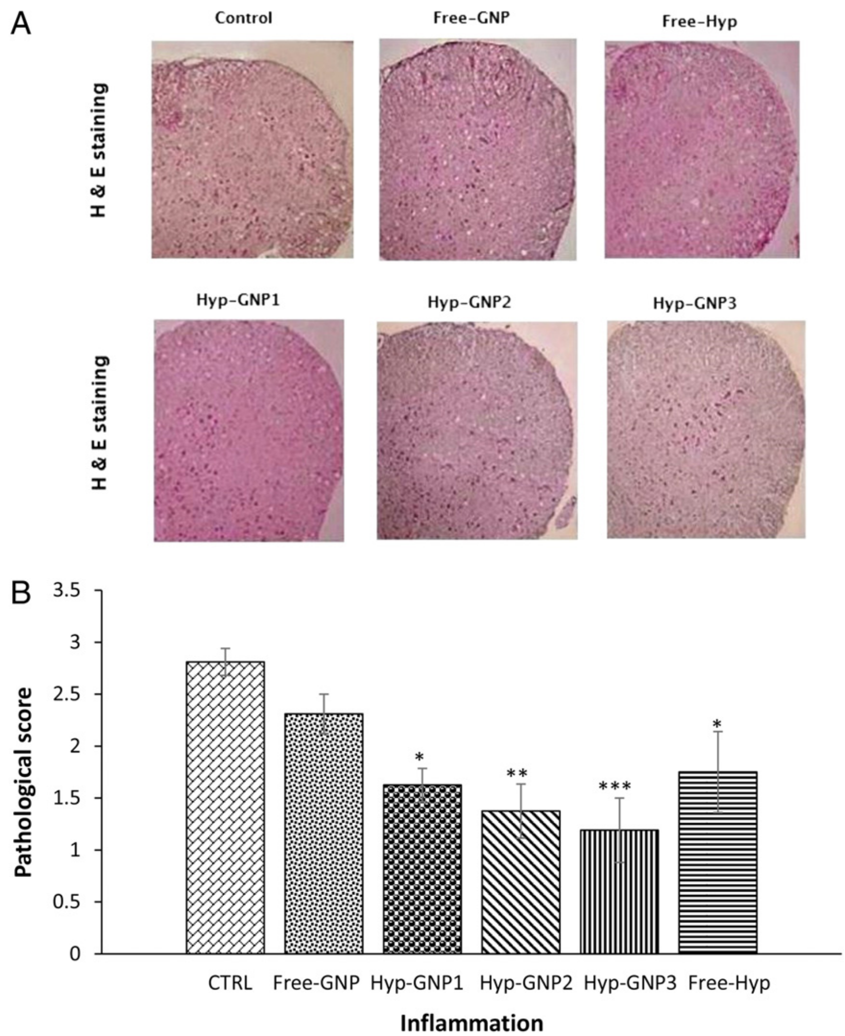

Figure 3. The inflammatory cell infiltration into the spinal cords after administration of Hyp-GNP and hyperforin. **(A)** The figure illustrates the reduced inflammatory cell infiltration into the spinal cords after treatment.**(B)** Histological features were scored semi-quantitatively as described in the Materials and methods.*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$ compared with vehicle treated control. Values are presented as mean ± SEM.

As a result, hyperforin and GNP added together have a synergistic effect on Th2 and Treg expansion. Moreover, delivery of hyperforin with nanoparticles (Hyp-GNP3) inhibited Th1 and Th17 differentiation more efficiently than did free-Hyp ($P < 0.05$).

Discussion

In the current research, we have provided a drug–gold nanoparticle conjugate (Hyp-GNP) to examine in experimental multiple sclerosis. The design of this Hyp-GNP model was due to gold nanoparticles (GNPs) have proved great promise in drug delivery due to excellent biocompatibility, low toxicity and immunogenicity,[25] simple conjugation chemistry and stability under most in vivo conditions.[26]

Therefore, the surface functionalization with drugs gives additional ability and enables GNPs as one of the best nanoparticles for biomedical applications.

Several studies have indicated that the pure components isolated from herbs have the reducing and stabilizing ability to provide a biocompatible GNP.[27] Hsieh et al., demonstrated that the combination of EGCG (the component of green tea) with GNP can inhibit bladder cancer in mice.[28] In order to improve pharmaceutical properties and therapeutic efficacy of hyperforin, we used GNPs directly as drug carriers, by loading hyperforin on the gold nanoparticle surface.

To the best of our knowledge, this is the first report in which hyperforin in combination with gold nanoparticle is used for EAE treatment.

A previous study in our lab demonstrated hyperforin dose relatedly inhibited the clinical course of EAE.[23] The present study further evaluated the in vivo and in vitro anti-inflammatory and immunomudulatory efficacy of hyperforin in combination with GNP.

Our study found mice treated with Hyp-GNP displayed significantly decreased mean clinical scores and delayed the onset of disease compared to mice receiving PBS, free-GNP, or free-Hyp.

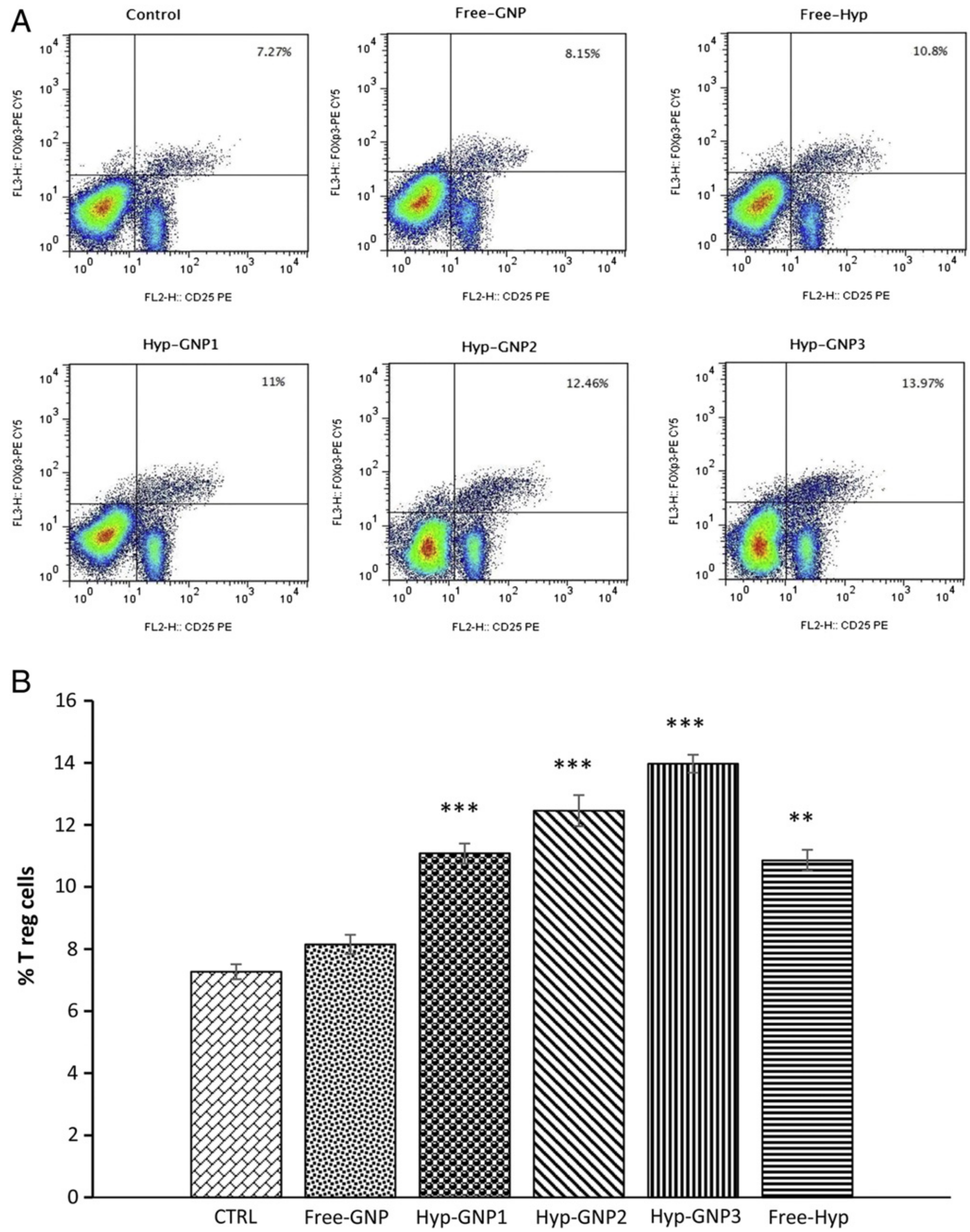

Figure 4. Hyp-GNP and hyperforin administration promoted the frequency of Treg cell population. **(A and B)** Splenocytes were stained with anti-CD4 FITC, anti-CD25 PE, then cells were fixed and permeabilized and stained with anti-Foxp3 PE-CY5. Lymphocytes were gated by forward and side scatter. Lymphocytes were then gated on the $CD4^+$ population and the quadrant marker was then set on CD25 PE vs. isotype control IgG2a PE-CY5. Results are expressed as mean ± SEM.***P* < 0.01, ****P* < 0.001 compared with vehicle control.

It is well known that the pathogenesis of MS is due to impairments of immune cells and polarization of the cells toward either Th1 or Th2 response can significantly influence autoimmunity.[29] We have shown that the Hyp-GNP products increase IL-4 and IL-10 secretion by Th2 cells. Our study also showed increased GATA-3 expression in murine splenocyte and the brain, hence, according to the results it appears that Hyp-GNP induces production of Th2 cytokines via up-regulation of GATA3 and subsequently Th differentiation to Th2 subsets.

In parallel with our results, Cabrelle et al., showed that treatment with hyperforin leads to suppressed production of IFN-γ and upregulation of GATA-3 in activated T cells.[30]

As a result, the present study showed that free-Hyp not only induces the Th2 differentiation, but also enhances the expression of GATA-3 and production of IL-4 and IL-10, which promotes the remission of the disease.

The effects of free-Hyp in our study are in agreement with the study reporting the suppressive effects of hyperforin on IFN-γ but

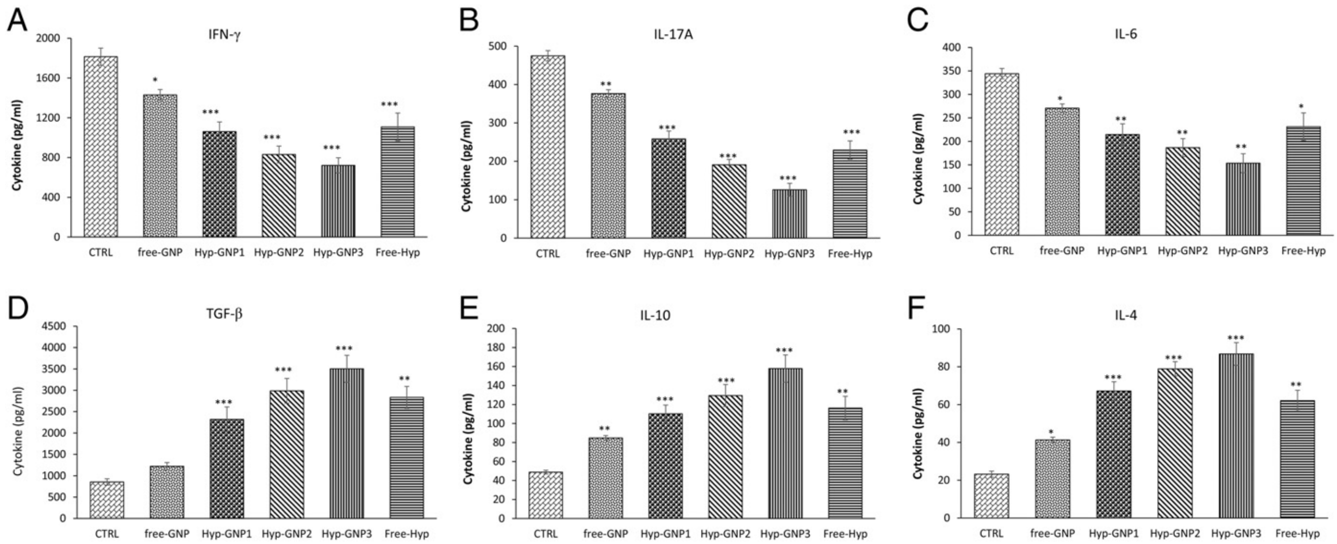

Figure 5. Effects of Hyp-GNPs and hyperforin treatment on pro-inflammatory and anti-inflammatory cytokines. The figure shows that Hyp-GNPs and hyperforin treatment led to suppression and enhancement of pro **(A-C)** and anti-inflammatory **(D-F)** cytokines, respectively. The values are means of triplicates at each point and the error bars represent SEM.Statistical analysis was performed by one-way ANOVA followed by post hoc Tukey's test.**P* < 0.05, ***P* < 0.01, ****P* < 0.001.

not IL-4.[30] Furthermore, the possible mechanism involved in disease progression is the expression of T-bet.[31] The attenuation of EAE by Hyp-GNPs and free-Hyp was also associated with the inhibition of T-bet expression and IFN-γ production. Our results demonstrated that Hyp-GNPs and free-Hyp lead to down regulation of T-bet and subsequently inhibition of pro-inflammatory cytokines (IFN-γ, IL-6 and IL-17A). The induction of IL-4/GATA3 in differentiated Th2 cells and the concomitant down-regulation of IFN-γ/T-bet in Th1 cells by hyperforin products identified these drugs as novel targets for MS therapy.

However, Th1/Th2 paradigm in EAE/MS has been challenged with identification of Th17 as well as Treg cells.[32]

Hyperforin inhibited the differentiation and activation of Th17 and Th1 cells and induced Treg cell differentiation. IL-17 is an important cytokine in autoimmune diseases, and Zhu et al., have recently shown inhibition of IL-17 protects against EAE.[33] A study by Tesmer et al., also suggests that IL-17 may play a more significant role in EAE than IFN-γ.[34] In this study, we demonstrated for the first time that Hyp-GNP inhibits the differentiation and effector function of Th17 cells more effectively than free-Hyp. It is appears that hyperforin suppressed Th17 differentiation and effector function via down regulation of ROR-γt. We, also, demonstrated that a lower dose of Hyp-GNP is required to inhibit IL-17 production by T cells than similar treatment with free-Hyp. There are no reports of the effect of Hyp-GNP on Th17 cells and IL-17 production, but the study by Cabrelle et al., demonstrated that hyperforin protected animals from EAE induction and this protection was associated with reduced IFN-γ production.[30] We have observed similar immunosuppression by free-Hyp and Hyp-GNP especially in the group treated with Hyp-GNP at a dose 5 times lower than that of free-Hyp.

T cell polarization is largely organized by cytokine-mediated signals.[35] TGF-β and IL-6 are two key cytokines that drive T cells to differentiate into Tregs or Th17 cells.[36] TGF-β alone causes Treg differentiation, whereas TGF-β in the presence of IL-6 induces Th17 differentiation.[37] It was evident in this study that the effect of free-Hyp and Hyp-GNPs on Th17 cells is dependent, in part, on IL-6. Our study revealed that free-Hyp or Hyp-GNP treatment increased TGF-β and reduced IL-6 production, which established a favorable milieu for Treg, but not Th17 cell differentiation.

Furthermore, we have found that there is an increase in $CD4^+$ $CD25^+Foxp3^+$ Treg cell numbers in free-Hyp or Hyp-GNPs treated mice and it has been suggested that this may be due to TGF-β in vivo. The decrease of IL-17A and increase of TGF-β production were further supported by significantly reduced expression of ROR-γt and Foxp3 upregulation. Thus, based on our findings, it is likely that free-Hyp or Hyp-GNPs suppresses EAE by inhibiting the differentiation and activity of Th17 and induction of Treg cells.

Additionally, the results revealed that although, the CDI and EAE scores were changed between free-GNPs and control groups and the levels of Th1 and 2 were significantly decreased and increased, respectively, but the pathological scores were not changed between the groups. Previous investigations demonstrated that GNP plays as an anti-inflammatory factor and, hence, can regulate immune responses.[38] According to our results, it can be concluded that GNP have a similar function and reduce inflammation via inhibition and activation of Th1 and Th2, respectively. The results from CDI and EAE scores also confirmed the conclusion. Based on the fact that GNP was unable to alter pathological scores, hence, it may be hypothesized that GNP regulates Th1 and Th2 functions in spite of infiltrations of the cells.

Histological examination showed Hyp-GNP suppresses clinical paralysis in EAE disease by reducing inflammatory cell infiltration into the spinal cord. Dell'Aica et al., have also

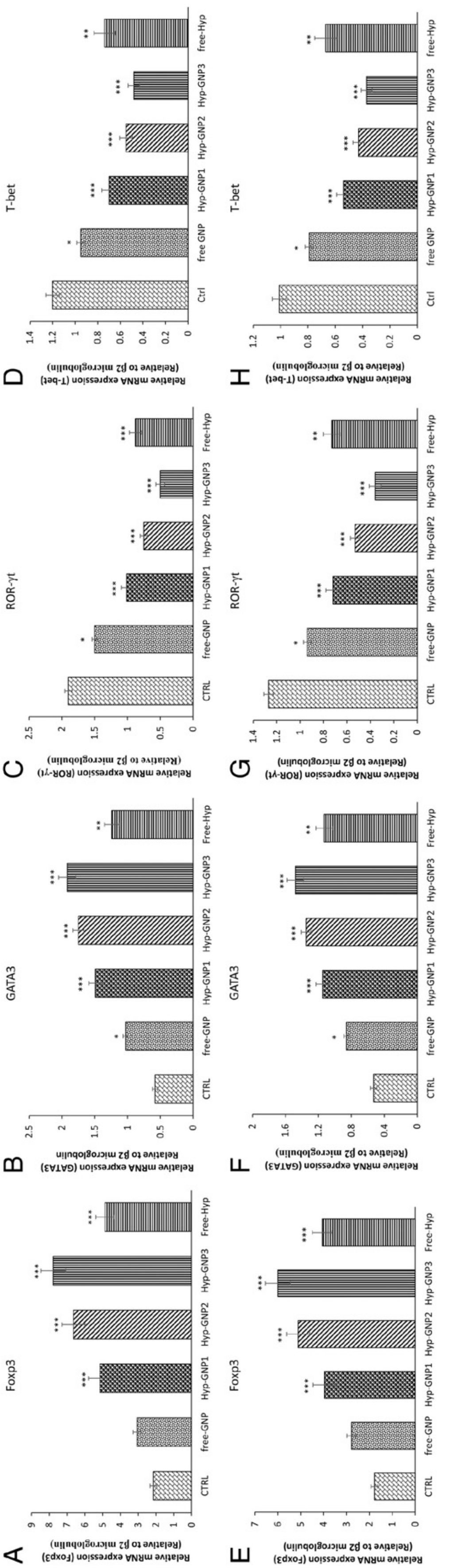

Figure 6. Expression of transcription factors (T-bet, GATA3, ROR-γt and Foxp3) in the spleen and brain of mice with EAE. Brain samples and splenocytes were isolated from mice on day 21 post immunization. Total RNA was extracted and reverse transcribed into cDNA. Equal quantities of cDNA from each sample analyzed by qPCR using β2 microglobulin as the internal control. mRNA expression levels of transcription factors in the spleen **(A-D)** and brain **(E-H)** of mice.Results are expressed as mean ± SEM.*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$ compared with vehicle control.

demonstrated that hyperforin reduces leukocyte infiltration to the CNS via down-regulation of chemokines and adhesion molecules.[39] Moreover, a substantial reduction of brain damage has also been reported by hyperforin in ischemic stroke and Alzheimer disease.[40,41] It seems that hyperforin plays as an anti-inflammatory agent through its direct effect on T cells by blocking their differentiation to Th1 cells as well as the inhibition of pro-inflammatory cytokine production and subsequent recruitment of inflammatory cells into the CNS.

Previous investigations revealed that hyperforin can modulate MAPK/AP-1-dependent pathway in human intestinal epithelial cells.[42] Another study demonstrated that hyperforin plays as an inhibitor of NF-κB.[43] Hyperforin also is able to inhibit synthesis of microsomal prostaglandin E2 (PGE2) in in vivo and in vitro conditions.[44] PGE2 is an pro-inflammatory molecule that plays key roles in the induction of EAE.[45] Based on the effects of hyperforin on the molecular pathways of the cell systems, hence, it may be concluded that the GNP-conjugated molecules can be more effective to modulate immune responses in EAE, as the MS animal model. Additionally, as mentioned in the previous sentences, hyperforin has anti-inflammatory functions in pro-inflammatory based diseases including ischemic stroke and Alzheimer disease.[40,41] Therefore, it may be proposed that hyperforin loaded gold nanoparticle can be considered as potential agent for treatment of MS.

Our study revealed that Hyp-GNP at very low doses was able to inhibit EAE progression, while similar results were observed with high dose of free-Hyp. Thus, Hyp-GNP offers several advantages over free-Hyp as a therapeutic agent that includes low dose Hyp-GNP leading to a lower toxicity, fewer side effects and significantly enhancing therapeutic activity of hyperforin.

In summary, Hyp-GNP and hyperforin significantly reduced the disease clinical score and infiltration of inflammatory cells into the CNS, especially by favoring Treg cells. Our finding indicated that Hyp-GNP and hyperforin treated mice showed an increased expression of Foxp3 and GATA3 versus decreased expression of T-bet and ROR-γt, demonstrating the immuno-suppressive mechanisms of compounds. Additionally, we showed Hyp-GNP was superior to an equal dose of free-Hyp in ameliorating disease severity, which indicated that hyperforin in conjugation with GNP, a novel new nano pharmaceutical, has synergistic immunomudulatory properties and may delay or halt the progression of EAE.

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.nano.2016.04.001.

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